

- 5 The curve C has polar equation $r = \frac{1}{\pi - \theta} - \frac{1}{\pi}$, where $0 \leq \theta \leq \frac{1}{2}\pi$.

(a) Sketch C .

[3]

- (b) Show that the area of the region bounded by the half-line $\theta = \frac{1}{2}\pi$ and C is $\frac{3 - 4 \ln 2}{4\pi}$. [6]

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